

Alternative Pathway to Phase Down Coal Power and Achieve Negative Emission in China

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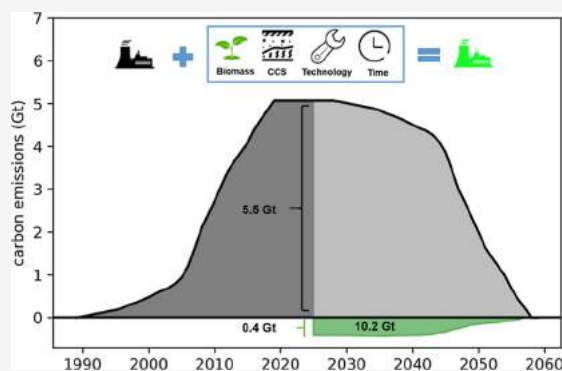
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ABSTRACT: Although widely recognized as the key to climate goals, coal “phase down” has long been argued for its side effects on energy security and social development. Retrofitting coal power units with biomass and coal co-firing with a carbon capture and storage approach provides an alternative way to avoid these side effects and make deep carbon dioxide emission cuts or even achieve negative emission. However, there is a lack of clear answers to how much the maximum emission reduction potential this approach can unlock, which is the key information to promote this technology on a large scale. Here, we focus on helping China’s 4536 coal power units make differentiated retrofit choices based on unit-level heterogeneity information and resource spatial matching results. We found that China’s coal power units have the potential to achieve 0.4 Gt of negative CO₂ emission in 2025, and the cumulative negative CO₂ emission would reach 10.32 Gt by 2060. To achieve negative CO₂ emission, the biomass resource amount should be 1.65 times the existing agricultural and forestry residues, and the biomass and coal co-firing ratio should exceed 70%. Coal power units should grasp their time window; otherwise, the maximum negative potential would decrease at a rate of 0.35 Gt per year.

KEYWORDS: coal phase down, negative emission potential, biomass and coal co-firing with CCS, unit-level analysis, economic costs



INTRODUCTION

The last-minute wording change from coal “phase out” to “phase down” in the COP26 climate conference reflects the struggle to both keep the 1.5 °C goal alive and to reduce the energy security and social development problems associated with coal phase down.¹ Compulsorily retiring the coal power units would cause many side-effects, such as huge stranded assets,^{2–4} massive job losses,⁵ power grid instability, and energy security risk.⁶ Therefore, the action on coal power exit should be designed carefully. In fact, simply phasing out coal is still inadequate to keep the temperature rise limit goals within reach. In the context of large-scale electrification in the future, many studies agree that the power sector would be required to achieve negative emission with the help of bio-energy with carbon capture storage (BECCS).^{7–15} Consequently, several studies have begun to explore the feasibility of achieving negative emission by retrofitting existing coal power units as biomass and coal co-firing power units with carbon capture and storage (BCP-CCS).^{16–21} By utilizing the existing infrastructure, this approach can prevent the original coal power units from forced shutdowns, avoid above-mentioned side-effects, and may help the coal power sector achieve net-zero or even negative emission. However, the answer to how much negative CO₂ emission potential this approach can

achieve in the coal power sector and the corresponding cost is still not clear. China generates over half of the coal power in the world, and the emission caused by coal power accounts for around 40% of the national CO₂ emission. We choose the coal power units in China as a case to answer this question.

The negative emission potential of BCP-CCS in the coal power units is largely decided by its detailed deployment pathway because large uncertainties come from the differentiated retrofit strategies of each unit, such as the biomass and coal co-firing ratio, transport amount and distance of the biomass resources and the captured carbon dioxide emission, and commissioning year and the retrofit time.^{22–28} Also, these strategies will also be influenced by the policy support in large-scale energy crop plantation, technology development breakthroughs, and the carbon market. Although most of the existing studies have acknowledged the importance of BECCS,

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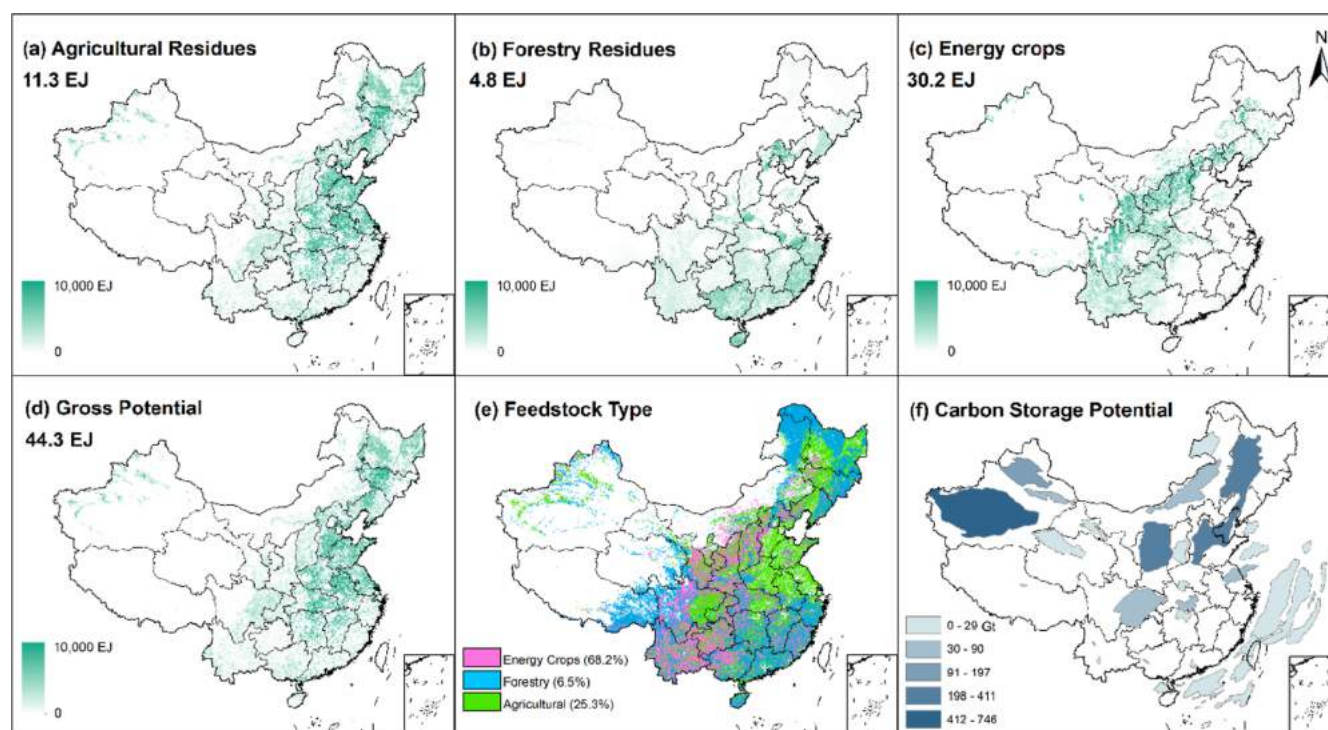


Figure 1. Spatial distribution of China's bioenergy potential and carbon storage potential. (a–c) represent the bioenergy spatial distribution map sourced from the existing agricultural residues, the existing forestry residues, and the future energy crops, respectively. (d) represents the gross bioenergy potential of the above-mentioned three feedstock types after optimal selection. (e) represents the feedstock type of (d). (f) represents the carbon capture and storage potential of the on-shore saline basin.

few have explored how to deploy BECCS taking into account the above-mentioned uncertainties. Lu et al. proved the feasibility to achieve negative emission through a sample set of an integrated gasification cycle system combined with carbon capture and storage from a theoretical perspective.²⁸ Xing et al. drew a marginal abatement cost curve for the coal power sector retrofitted with BECCS based on county-level data with spatial explicit methods.²⁹ They both advanced the understanding of the deployment of BECCS in China. However, limited by the data and the research scope, they are unable to tell how each unit would make optimal BECCS retrofit decision. Meanwhile, the commissioning year and start- and finish-retrofitting time would largely influence the cumulative negative emission and life cycle economic cost.^{30–32} Therefore, the lack of unit-level analysis may hinder our understanding of the more realistic mitigation potential and the economic cost.

To address the above-mentioned research gaps, this study aims to answer how much mitigation potential these coal power units can achieve and at what costs; the following three specific sub-questions have been answered: (1) what is the maximum supply potential of biomass resources and how are they distributed in China? (2) How and at what cost would each coal power unit make its retrofitting decisions to realize the maximum emission reduction potential? (3) What are the overall mitigation potential and cost in the coal power sector and how would this potential be affected by key influential factors? The answers to these questions have important policy implications and deployment instructions for China's carbon neutrality pathway. The potential of negative CO₂ emission in China's coal power sector could provide insights into how much the other hard-to-abate sectors (such as freight transport and residential household sectors in the China context) could

emit in the future to ensure the realization of carbon neutrality goals (Figure S1).

METHODOLOGY

Biomass Resource Potential Assessment. The land protection and food security policy in China does not allow arbitrary conversion of cropland and forest land to other land use types, which confines the scopes of feedstock types to agricultural residues, forestry residues, and energy crops on marginal land. Also, it is precisely because of this national condition that a large amount of emission caused by land use change can be avoided.^{27,33–35} Agricultural residues mainly originate from the inedible stems of maize, wheat, rice, and so forth.^{36,37} Forestry residues are mainly collected from forest harvesting, wood processing, and forest clearing.^{38,39} By combining statistics data and the parameters reflecting the ratio and relationship between the residues and product,^{40,41} we calculated the provincial results. With the help of the Geography Information Science (GIS) method, we down-scaled the provincial result into a 1 km grid using net primary productivity data⁴² and the land used type dataset⁴³ as weight and allocation area (Section S1).

For energy crop assessment, the location-specific yield data for different types of energy crops in China were extracted from Li et al.^{44,45} and had been matched with the location of marginal land identified in this study according to the definition of Zhang et al.⁴⁶ which considered the land use type, natural nature reserves, and slope (Section S2). Due to the small number of energy crop trial samples, large-scale cultivation and management might arouse ecological risk. In this study, we only considered the yield of energy crops in the rain-fed scenario.

To avoid the same land used to grow different types of plants, we converted the potential of biomass resources into the heat value by assuming that the heat value of agricultural residues, forestry residues, and energy crops is 14.7, 17.3, and 16.3 GJ/t, respectively.^{47,48} Also, the heat value distribution maps are shown in Figure 1a–c. Then, we selected the maximum value in each grid to obtain the maximum bioenergy spatial distribution map in China, which can be found in Figure 1d. The feedstock sources corresponding to the heat value are shown in Figure 1e. This dataset has high spatial resolution and shows the largest biomass resource potential in China, which is achievable under the present natural condition (the workflow of bioenergy potential assessment is shown in Figure S2).

Coal Power Unit Dataset and the Spatial Matching of the Supply Chain (Biomass Resources—Coal Power Units—Carbon Storage Sites). The unit-level coal power dataset used in this study included 4536 units in service in 2018, with a total installed capacity of 990.6 GW, and the time span of each unit's commissioning is covered from 1989 to 2018.^{49,50} This dataset is built by integrating multiple data sources and covers more than 98% of the total coal power capacity in 2018.⁵¹ In this study, we only discussed existing operational coal power units, without considering planned units (Figures S3 and S4).

For achieving spatial matching, first, we divided the biomass resources map into a 10 km*10 km grid and considered the centroid as the biomass collection point (BCP). In order to avoid the same BCP shared by many power units while this BCP cannot provide enough biomass resources at the same time, we used the Thiessen polygon to divide the whole area into pieces, which means that each BCP serves the nearest coal power unit, and the total production of these backup BCPs determines the upper limit of each power unit's co-firing ratio. The carbon storage basins are mainly distributed in Northwest and Central China. The distance between the coal power units and the carbon storage points will differentiate the transportation cost of captured carbon dioxide. Therefore, we collected the on-shore saline basin dataset developed by Li et al. as our targeted storage area.⁵² Then, we broke these areas by county-level administrative data. The centroid points of each fragmented piece were set as potential carbon storage points. The ratio of each fragmented piece to the original area was used as a weight to allocate the storage basin's capacity. Then, we used the k-mean method to cluster these potential carbon storage points and constructed a Delaunay triangle network for GIS network analysis. In the end, we used the Closest Facility tool in ArcMap to help each unit find its nearest storage site and calculated the transportation distance of captured carbon dioxide⁵³ (Figure S5).

In the framework of the spatial matching, the biomass resource potential dataset, coal power unit dataset, and CCS storage dataset are used as data input. The output of this framework includes the upper bound of the biomass supply amount (it determines the maximum co-firing ratio), transportation distance to each BCP, and the transportation distance to the nearest CCS storage site. These results would be used as constraints and variables guiding each coal power unit to take retrofit action (Figure S6).

Scenario Setting. There are three key factors that would influence the maximum CO₂ emission reduction potential in China's coal power sector, which are the amount of biomass resource potential, the maximum co-firing ratio of biomass

with coal, and the time to start and finish co-firing retrofitting. For biomass resource potential, China has already paid attention to the utilization of agricultural and forestry residues, while this potential has not been completely exploited yet. Besides, the research on energy crops, such as miscanthus and switchgrass, is being promoted and advanced, but large-scale plantation confronts many economic and ecological uncertainties. Based on this background, two scenarios concerning biomass resource potential have been set up here: (1) current viable scenario (fully utilizing existing agricultural and forestry residues) and (2) future exploitable scenario (utilize residue resources and large-scale plantation of energy crops on marginal land). For technology breakthroughs in the biomass and coal co-firing ratio, the maximum technically feasible co-firing ratio is 20% in China, and the most existing co-firing ratios are below 5% and rarely exceed 10%, while globally, this co-firing ratio can reach over 96%.⁵⁴ Therefore, two scenarios concerning technology breakthroughs have been set up here: (1) the maximum co-firing ratio of biomass and coal co-firing is below 20%; (2) the maximum biomass and coal co-firing ratio can reach 100%. By combining the scenarios of the biomass resource potential and co-firing ratio, we can obtain four CO₂ emission reduction potential scenarios (Table 1). A

Table 1. Four Main Scenario Settings for Maximum Negative CO₂ Emission Potential Estimation^a

	maximum technical capability at present (co-firing ratio ≤20%)	great breakthrough of technology in future (co-firing ratio ≤100%)
the largest available biomass resources at present (A + F)	B–T–	B–T+
the largest biomass resources in the future (A + F + E)	B+T–	B+T+

^aNote: A represents agricultural residues, F represents forestry residues, E represents energy crops, and the co-firing ratio represents biomass and coal co-firing ratio.

business-as-usual (BAU) scenario is also set to compare with the above-mentioned scenarios, in which the coal power units operate without taking any retrofitting action. Besides, the time to start BCP-CCS retrofitting is also a key factor for maximum CO₂ emission reduction potential. Restricted by a life time of 40 years that we assumed for each unit, these power units will gradually retire naturally. According to our dataset, most existing coal power units will retire after 2045. For assessing the maximum emission reduction potential, we assume that all coal power units start and finish retrofitting in 2025 in these four scenarios. As for how much CO₂ emission reduction potential would be reduced if we delay our retrofit action, we will discuss this in the Sensitivity Analysis section.

Life Cycle Assessment of Carbon Dioxide Emission. As biomass growth will absorb carbon dioxide while the transportation process will cause extra carbon emission, CO₂ emission of coal power unit retrofitted with BCP-CCS technology should be calculated from the life cycle. To unify the evaluation framework and make results comparable, the CO₂ emission of coal power units should also include these processes, which are shown in eq 1.

$$\text{Emiss}_{\text{pp}} = \alpha \times (e_{\text{p1}} + e_{\text{p2}} + e_{\text{p3}}) \times C \times H \quad (1)$$

where $E_{\text{miss,pp}}$ (metric tonne) represents the life cycle CO_2 emission of the coal power unit; e_{p1} , e_{p2} , and e_{p3} represent the parameter of coal mining, transportation, and terminal combustion, with a specific value of 275.1, 14.8, and 801.7 g/kW h, respectively;⁵⁵ α represents the unit conversion factor (10^{-6}); C (kW) represents the installed capacity of the coal power unit; and H represents the operating hours.

The CO_2 emission of coal power units retrofitted with BCP-CCS mainly includes CO_2 absorption during biomass growth, CO_2 emission in the supply chain, pretreatment before combustion at the terminal, and carbon leakage after the combustion. To simplify the calculation process, it can be divided into three parts (eq 2): (1) We can combine the calculation of the CO_2 absorption amount during biomass growth and biofuel combustion at the terminal. This combination means that the CO_2 absorption of biomass will be stored underground by CCS technology (eq 3). (2) CO_2 emission in the process of collection, transportation, storage, and pretreatment (eq 4). (3) When biomass and coal are combusted at the power unit, the leakage of CO_2 emission is brought about by the coal combustion and coal mining and transportation process (eq 5). The leakage of CO_2 emission brought about by biomass combustion has been considered in eq 3.

$$E_{\text{BECCS}} = -E_1 + E_2 + E_3 \quad (2)$$

where the detailed calculation processes of E_1 , E_2 , and E_3 can be found in the following equation.

$$E_1 = \alpha \times e_{b1} \times C \times H \times r_b \times e_{\text{ccs}} \quad (3)$$

where e_{b1} represents the CO_2 absorption amount by unit biomass electricity (the value is 375.7 g/kW h);^{56,57} r_b represents the biomass and coal co-firing ratio; and e_{ccs} represents the carbon capture rate of CCS technology, which is assumed to be 90%. Other parameters and annotations are the same as those in the above-mentioned equations.

$$E_2 = \alpha \times (e_{b2} + e_{b3} + e_{b4} + e_{b5}) \times \sum_{j=1}^m k \times x_{ij} + \alpha \times e_{b6} \times \sum_{j=1}^m k \times x_{ij} \times d_{ij} \quad (4)$$

where e_{b2} , e_{b3} , e_{b4} , and e_{b5} represent the CO_2 emission by unit biomass electricity in the process of field package, pretreatment before storage, transfer and storage at the BCP, and pretreatment in the power unit, respectively. The specific value of these parameters is 2.36, 1.1, 0.295, and 2.457 g/kW h, respectively.⁵⁵ k represents the unit conversion factor of biomass production to electricity (the value is 4444.8 kW h/t). To simplify the calculation, we assume that 1 metric tonne of dry biomass resources can generate 16 GJ, and 1 GJ equals 277.8 kW h. e_{b6} represents the emission that transports one unit of biofuel for 1 km, which equals 0.009 g/(kW h*km). x_{ij} (metric tonne) represents the biomass resource amount transported from BCP j to coal power unit i . d_{ij} (km) represents the transportation distance between BCP j and power plant i ; other parameters and annotations are the same as those in the above-mentioned equations.

$$E_3 = (1 - r_b) \times \alpha \times \{e_{p1} + e_{p2} + (e_{p3} \cdot (1 - e_{\text{ccs}}))\} \cdot C \cdot H \quad (5)$$

where r_b represents the biomass and coal co-firing ratio and e_{ccs} represents the carbon capture rate. α , e_{p1} , e_{p2} , e_{p3} , C , and H are the same as those in the above-mentioned equations.

Life Cycle Assessment of Levelized Cost of Electricity.

The levelized cost of electricity (LCOE) is used to measure the average net present cost per unit of power generation in the power unit's whole life cycle from commissioning to retirement. It is widely used to compare the economic cost of different power generation technologies. The items in this study mainly include investment cost, operation and management cost, fuel cost, and depreciation cost. As we want to describe the differentiation between each unit, more details, such as the retrofit year, transportation distance and the biomass resource amount, carbon transportation distance captured by CCS technology, and the co-firing ratio of coal and biomass, are taken into consideration. The calculation equations are shown below. To ensure the scientific and reasonable selection of the parameters, we made a unified parameter framework based on the model. More details on the process of parameter collection, selection, and references are shown in Section S4 and Tables S4–S7.

$$L_{\text{BECCS}} = \frac{\sum_{t=1}^n [\beta_1 \times C + (\beta_2 + \beta_3) \times C + \alpha \times \beta_4 \times C \times H - \beta_5 \times \beta_1 \times C] + L_{\text{add-be}} + L_{\text{add-ccs}}}{\sum_{t=1}^n E \times (1 + r)^{-t}} \quad (6)$$

where L_{BECCS} (\$/kW h) represents the levelized cost of BECCS, $L_{\text{add-be}}$ (\$/kW h) and $L_{\text{add-ccs}}$ (\$/kW h) represent the increased cost as being retrofitted with CCS and biomass and coal co-firing technology, and n represents the life span of the unit. We assume that each coal power unit can operate for 40 years in this research; β_1 represents the investment cost of unit install capacity (\$3274/kW). β_2 and β_3 represent the fixed and variable operation and management costs, with a value of \$41.66/kW and \$31.45/kW, respectively; β_4 represents fuel cost per unit of coal power generation (\$11.45/MW h); and β_5 represents the depreciation rate. We assume that the depreciation cost is 5% of the investment cost; C , H , α , E , and r are the same as those in the equations mentioned above.

$$L_{\text{add-be}} = \frac{\sum_{t=T}^n [\gamma_1 \times C + (\gamma_2 + \gamma_3) \times C + F - \beta_5 \times \gamma_1 \times C]}{\sum_{t=1}^n E \times (1 + r)^{-t}} \quad (7)$$

where T represents the time to retrofit; γ_1 represents the investment cost of retrofitting a unit as biomass and coal co-firing technology (\$327.2/kW); γ_2 and γ_3 represent the increased fixed and variable operation and maintenance (O&M) cost per unit installed capacity, with a value of \$18.85/kW and \$27.5/kW, respectively; and F represents the fuel cost of coal and biomass.

$$F = \alpha \times [\beta_1 \times C \times H \times (1 - r_b) + (\gamma_4 + \gamma_5 + \gamma_6) \times \sum_{j=1}^m k \times x_{ij} + \gamma_7 \times \sum_{j=1}^m (k \times x_{ij} \times d_{ij})] \quad (8)$$

where γ_4 , γ_5 , γ_6 , and γ_7 represent the cost parameters of biofuel collection, pretreatment, and transportation, with a value of \$10.3/MW h, \$6.15/MW h, \$6.75/MW h, and \$6.75/(MW h*km), respectively and r_b , k , x_{ij} , d_{ij} , β , and α are the same as those mentioned above.

$$L_{\text{add-ccs}} = \frac{\sum_{t=T}^n [\varepsilon_1 \times C + \text{Emiss}_{\text{pp}} \times (\varepsilon_2 + \varepsilon_3) + \text{Emiss}_{\text{pp}} \times d_{\text{ccs}} \times \varepsilon_4 - \beta_5 \times \varepsilon_1 \times C]}{\sum_{t=1}^n E \times (1 + r)^{-t}} \quad (9)$$

where ε_1 represents the increased investment cost of CCS per unit installed capacity (\$3274/kW); ε_2 and ε_3 represent fixed and variable O&M cost per unit captured CO₂ emission, with a value of \$40/t CO₂ and \$17.39/t CO₂, respectively; ε_4 represents the cost of transporting one tonne of carbon dioxide [\$0.1195/(t*km)]; and d_{ccs} (km) represents the transportation distance between the coal power unit and the carbon storage site. Emiss_{pp} , β_5 , E , and r are the same as those mentioned above.

$$T = R_y - O_y \quad (10)$$

where T represents the time to retrofit, R_y represents the retrofit year, and O_y represents the commissioning year of the unit.

Unit Carbon Dioxide Reduction Cost. Unit carbon dioxide reduction cost is used as the cost-effectiveness analysis for different retrofit technologies, which represents the cost of reducing one unit of CO₂ emission.

$$C = \frac{(L_r - L_o) \times C \times H}{E_r - E_o} \quad (11)$$

where C (\$/tonne) represents unit CO₂ reduction cost; L_o and L_r (\$/kW h) represent the economic cost of the coal power unit and the retrofitted unit, respectively; and E_o and E_r (metric tonne) represent the CO₂ emission of the coal power unit and the retrofitted unit, respectively. Other annotations are the same as those mentioned above.

Annual and Cumulative Emission Reduction Potential. Based on the CO₂ emission calculation method mentioned above, we set a 10% co-firing ratio as a step and calculated the CO₂ emission reduction of each unit retrofitted with BCP-CCS from a co-firing ratio of 10–100%. It is worth noting that the CO₂ emission reduction amount does not always increase with the co-firing ratio; it will depend on the transportation distance, biomass amount, and so on. Therefore, we help each unit choose a co-firing ratio that can achieve maximum CO₂ emission reduction. In the end, taking the retirement year of each unit into consideration, we can calculate the cumulative CO₂ emission in each scenario.

RESULTS

Potential of Biomass Resource Supply and Carbon Storage in China. In our dataset, the total amount of agricultural residues is 0.8 Gt, equivalent to 11.3 EJ, which are mainly distributed in the North China Plains and Northeast China Plains (Figure 1a). Heilongjiang, Henan, and Shandong provinces have the largest production. As for forestry residues,

the total production of forestry residues in China is 0.3 Gt, equivalent to 4.8 EJ, mainly distributed in Southwest and Northeast China (Figure 1b). These biomass resources are based on the existing resources without additional planting. As for energy crops, the production of energy crops is 1.7 Gt on marginal land, which is equivalent to 30.2 EJ (Figure 1c). In the gross production scenario, the total production would be 44.3 EJ (Figure 1d), of which 25% comes from agricultural residues, 6% from forestry residues, and 68% from energy crops (Figure 1e). The energy intensity of agricultural residues is high in the north and that of forestry is high in the northeast and southeast. Energy crops have the potential to produce more energy in Southwest and Central China. From the perspective of the provincial-level total bioenergy potential, Yunnan, Sichuan, and Guangxi are the top three provinces with the maximum potential, which can provide 13.3, 8.5, and 6.8% of country-wide potential, respectively (literature review on biomass resource assessment can be found in Figure S7). As for the carbon capture and storage amount, the total amount of carbon dioxide storage potential is 3088 Gt in the onshore and the offshore basin, among which saline aquifers account for 75% of the total amount (Figure 1f). The Tarim Basin, the Bohai Bay Basin, and the Songliao Basin have the most abundant storage potential with 745.8, 411.3, and 272.7 Gt, respectively.⁵⁸

Maximum Negative Emission Potential of China's Coal Power Sector. If China's coal power units do not make any retrofit and operate as BAU, these coal power units would emit 5.5 Gt in 2025 when taking a life cycle perspective where the processes of coal mining and processing, transport and stock, and coal burning are included (as shown in the black line of Figure 2). However, if they make optimal retrofits under

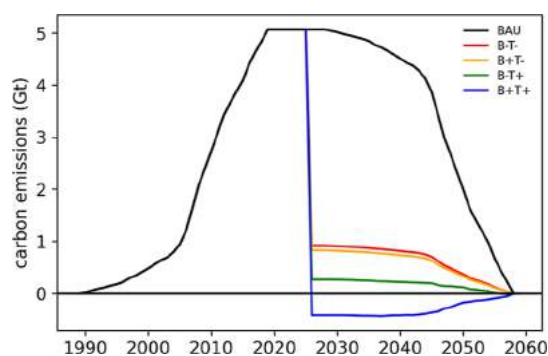


Figure 2. Carbon dioxide emission curve of China's coal power units in the past and the future scenarios.

the assumptions of four scenarios in Table 1, we find that the B–T– scenario (or the most resource-limited and technology-limited scenario) can help reduce 4.6 Gt of CO₂ emission in 2025, which is a big amount equivalent to 84% of emission in the BAU scenario; the B+T– and B–T+ scenario can help reduce 4.7 and 4.8 Gt of CO₂ emission in 2025, respectively. Only the B+T+ scenario can achieve negative CO₂ emission. This scenario can achieve 5.9 Gt of CO₂ emission reduction potential when compared with the BAU scenario, therefore achieving 0.4 Gt of negative CO₂ emission potential in 2025.

Besides, different from other negative emission technologies such as afforestation and reforestation, these coal power units, once retrofitted with BCP-CCS, can avoid or remove carbon dioxide stably in the long term, solving the carbon “locking-in”

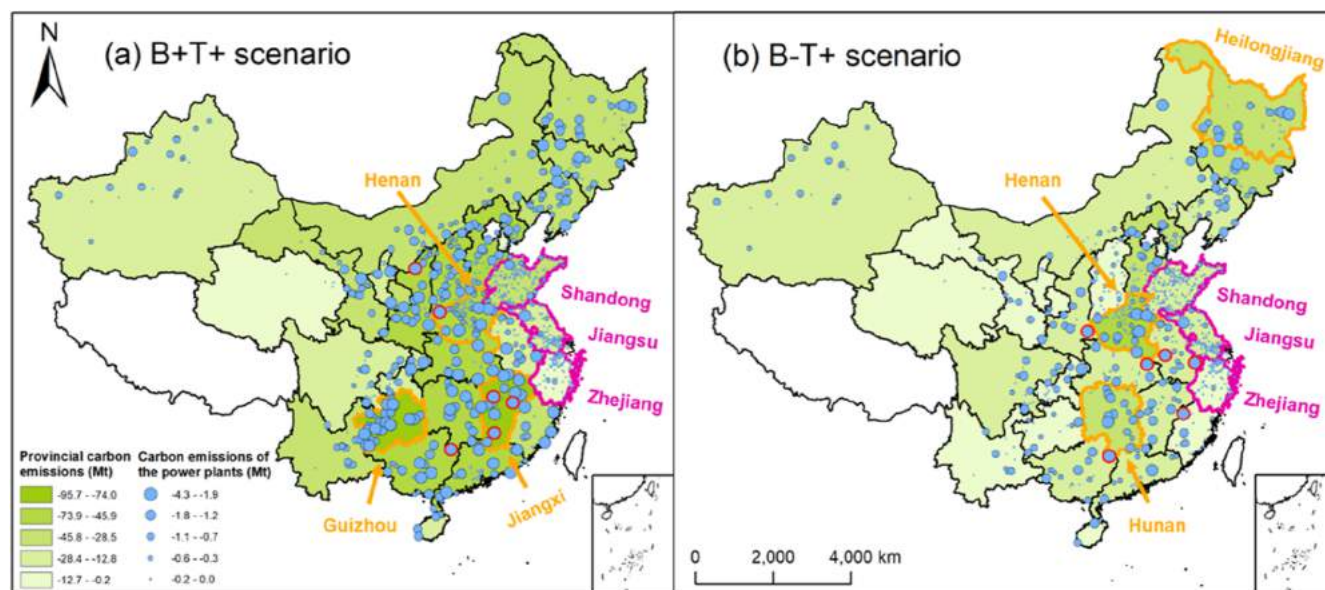


Figure 3. Negative carbon dioxide emission potential of China's coal power units and their spatial distribution map.

problem of existing infrastructure. Therefore, based on the information of the commissioning year and life span for each coal power unit, we calculate their cumulative emission and the cumulative emission reduction potential. From 1989 to 2020, coal power units in China emitted 58.1 Gt of carbon dioxide cumulatively under life cycle assessment. If we continue the BAU pathway, these coal power units would continue to produce CO₂ emission (or lock in carbon space) as much as 139.5 Gt accumulatively from 2020 to 2060. However, if we start and finish retrofitting in 2025 for all coal power units, the B+T+ scenario can bring about 10.3 Gt of the cumulative negative CO₂ emission space from 2025 to 2060, which is equivalent to nearly 1 year of China's total fossil carbon dioxide emission (9.8 Gt total fossil fuel-related emission in 2020 according to the calculation of the Climate Action Tracker). The other three scenarios, B-T+, B+T-, and B-T+, can reduce cumulatively 108.4, 95.5, and 93.5 Gt of CO₂ emission during the same period, respectively.

Negative Emission Units and Retrofit Economic Cost.

Due to the spatial heterogeneity of the 4536 coal power units considered in this study, a unit-by-unit cost-effectiveness assessment can better prove the economic feasibility of BCP-CCS and facilitate large-scale deployment. Based on the above-mentioned unit-level optimization results, we can estimate the emission reduction potential of each unit (or the total amount of negative emission it can possibly achieve). At the unit level, only part of the coal power units in B+T+ and B-T+ scenarios can achieve negative emission. Calculations for all scenarios show that for coal power units to achieve negative CO₂ emission, the co-firing ratio should at least exceed 30%. Therefore, based on the currently highest possible co-firing ratio in China (20%), no coal power units with BCP-CCS would be able to achieve negative emission (or remove carbon dioxide from the atmosphere). In the B+T+ and B-T+ scenarios, the spatial distribution maps of coal power units with the potential to achieve negative emission are shown in Figure 3. In the B+T+ scenario, 2468 coal power units can achieve negative emission (Figure 3a). Most of the coal power units would be concentrated in Shandong, Jiangsu, and Zhejiang provinces, where 430, 224, and 170 coal power

units in the respective province can achieve a negative emission of 28.5, 14.9, and 6.1 Mt in 2025, respectively. From the perspective of provincial-level potential of negative emission, Guizhou, Jiangxi, and Henan provinces would be the top three provinces, with the annual negative CO₂ emission potentials reaching 95.7, 74.0, and 65.2 Mt, respectively. The top 10 coal power units with the largest CO₂ emission potential are distributed in Jiangxi, Guangxi, Shaanxi, and Henan provinces, among which one individual coal power unit can bring about as much as 3.11–4.35 Mt negative CO₂ emission (Table S3). In the B-T+ scenario, 2082 coal power units can achieve negative CO₂ emission (Figure 3b). Most of the coal power units concentrate in Shandong, Jiangsu, and Zhejiang provinces, where 416, 217, and 181 power units in the respective province can achieve a negative emission of 22.6, 14.0, and 5.3 Mt in 2025, respectively. From the perspective of provincial-level potential of negative emission, Henan, Heilongjiang, and Hunan would be the top three provinces with the annual negative CO₂ emission potential reaching 49.1, 34.1, and 32.8 Mt, respectively. The top 10 coal power units with the largest CO₂ emission potential are distributed in Anhui province (Table S3), among which one individual coal power unit can bring about as much as 0.65–2.47 Mt negative CO₂ emission.

In terms of economic cost, if the nationwide coal power units are retrofitted with BCP-CCS technology, the additional economic cost would be between 278.0 and 278.4 billion dollars in all four scenarios. As equipment retrofitting cost is expensive, biofuel cost takes up a relatively smaller proportion, and the data distribution of LCOE in the four scenarios is similar. The costs of 25–75% of coal power units are in the range of 0.091 to 0.117 USD/kW h, and the median value is 0.1 USD/kW h. Compared with other low-carbon generation technologies, such as wind and solar with the value of LCOE between 0.054 and 0.087 USD/kW h,⁵⁹ BCP-CCS would only be competitive under the policy support or economic incentives.

However, it is worth noting that BCP-CCS is the only technology that can help coal power units achieve negative emission. Also, it plays an important role in unlocking the CO₂

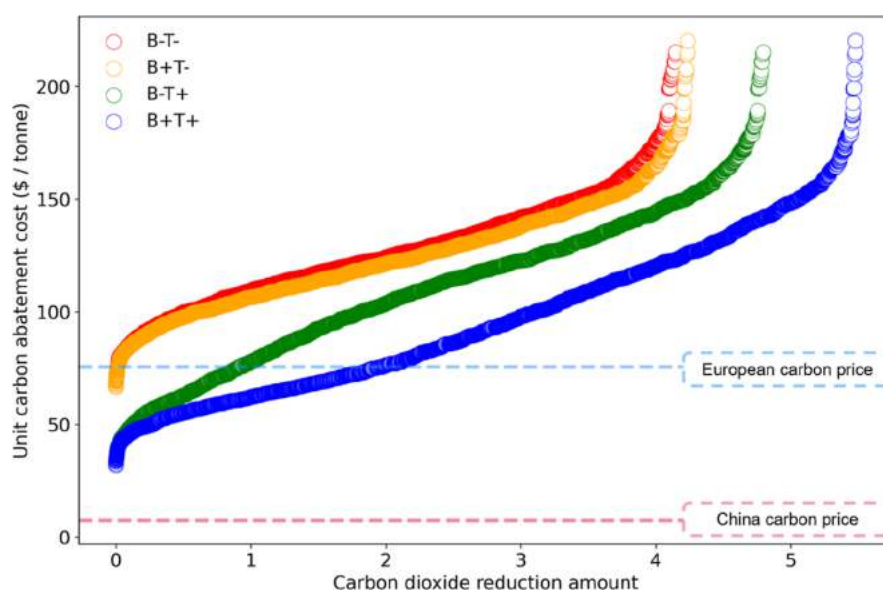


Figure 4. Carbon dioxide emission reduction curve of four major scenarios.

emission space, especially for developing countries that are still building new power units. According to the CO₂ emission reduction amount and the added retrofitted cost of each coal power unit, we can calculate the unit CO₂ emission reduction cost. By ranking the unit CO₂ emission reduction cost in the ascending order and summing each unit's emission with the emission amount of the preceding units, we can draw the marginal emission reduction cost curve of BCP-CCS technology (Figure 4), which may provide some information and a basis for carbon price setting. To achieve a 1 Gt emission reduction amount, the last-unit CO₂ emission reduction cost is 109.5, 106.4, 77.7, and 62.1 USD/t CO₂ in B-T-, B+T-, B-T+, and B+T+, respectively. China launched a carbon market on July 16 in 2021. As of March 31, the average carbon market price is around 7.5 USD/t CO₂. Also, the carbon price in the same period in Europe is 75.5 USD/t CO₂.⁶⁰ Under the current carbon price in China, no power unit would choose CBP-CCS from the perspective of profitability, while under the European carbon price, around 90–1870 (1.9–41%) coal power units would choose to achieve low carbon transformation. Also, the total CO₂ emission reduction amounts of the B-T+, B+T-, B-T+, and B+T+ scenario under this carbon price are 8.8, 12.5, 911.4, and 1971.8 Mt, respectively.

Sensitivity Analysis. As mentioned before, the amount of biomass resource potential, the maximum co-firing ratio of biomass with coal, and the time to start and finish co-firing retrofitting are the three key factors that would have great impacts on the emission reduction potential results. The above-mentioned four scenarios represent the maximum CO₂ emission reduction potential under four different specific assumptions. To further explore how these three factors affect the results and under what circumstances it would be impossible to achieve net-zero, we performed a sensitivity analysis on these three factors.

First, we fixed the time to start and finish co-firing and only considered changes in biomass resource potentials and the maximum co-firing ratio. Biomass resources are reduced at a 10% rate from the potential of biomass resources in the B+T+ scenario, while the co-firing ratio is reduced at the 10% rate from 100%. According to our results shown in Figure 5a, only

when biomass resources were higher than 60% of the potential in the B+ scenario (26.6 EJ), the overall coal power generation sector in China could start to achieve negative emission. The existing agricultural and forestry residues cannot meet the demand of BCP-CCS retrofitting, and 0.59 Gt energy crops (21.5% of the potential energy output from energy crops) would be needed additionally. In terms of the maximum co-firing ratio, the co-firing ratio should exceed 70% for the whole sector starting to achieve the negative emission, which is much higher than the 30% co-firing ratio mentioned above for a few coal power units to achieve negative emission. Then, we let the biomass resource potential and the maximum co-firing ratios change simultaneously. Here, we found that if the maximum co-firing ratio can only reach 70%, biomass resources should provide at least 39.9 EJ of energy to achieve negative emission. The annual total negative emission potential would be 349 Mt in 2025, which can offset France's CO₂ emission in 1 year. If the biomass resources can only provide 26.6 EJ of energy (60% of the potential in the B+T+ scenario), the co-firing ratio would need to reach 100% to realize negative emission, and the consequent negative emission potential would be 502 Mt.

The time to start and finish co-firing retrofitting will also affect the amount of cumulative emission reduction potential, which is essential to keep the temperature limit goals within reach. Our data shows that from 2045, a large number of coal power units would reach the end of their lifetime and need to be decommissioned. In 2050, the total capacity of coal power units would only account for 40% of the current level in 2018 assuming that they would retire naturally. Therefore, it makes no sense to further retrofit these retiring units, implying a narrower window to achieving the potential of realizing negative emission in the coal power sector. To quantify the effect of the start-to-retrofit time, we calculated how much cumulative emission reduction potential would be lost for every 5 years of postponing the start-to-retrofit time. An extra set of scenarios that represent different combinations of biomass resource potential and the maximum co-firing ratio has been set up. All these 11 scenarios can meet the lowest thresholds (i.e., at least 60% of the potential in the B+T+ scenario and at least 70% of the co-firing ratio) to achieve

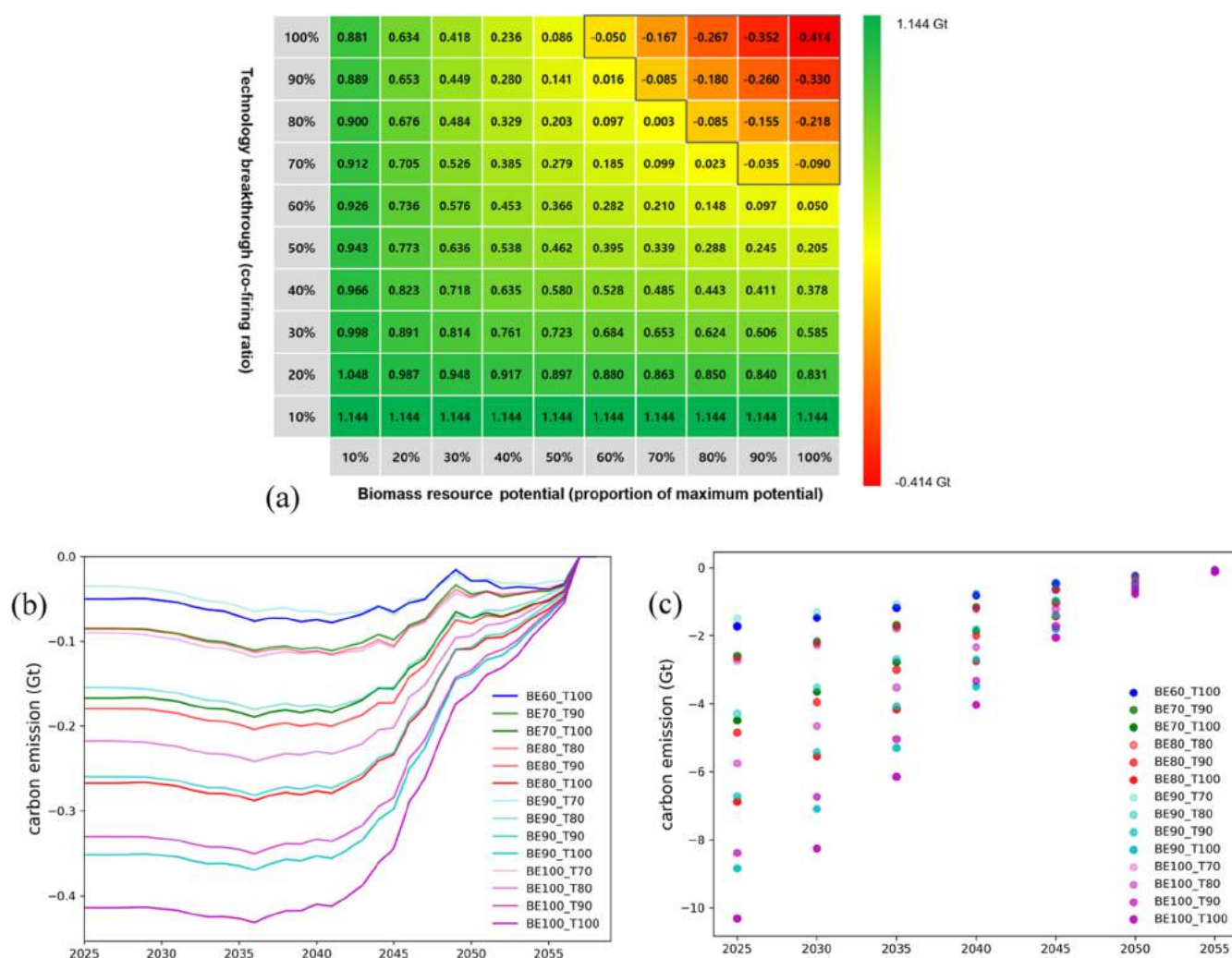


Figure 5. Sensitivity analysis of negative emission potential retrofitted with BCP-CCS affected by biomass resource potential, technology breakthrough, and start-to-retrofit time. (a) indicates the annual CO₂ emission amount affected by the biomass resource potential and co-firing ratio if all the units are retrofitted in 2025. (b) indicates the annual CO₂ emission curve if the power sector starts and finishes retrofitting in 2025 in the 11 negative emission scenarios selected according to (a). The area enclosed by the curve and the zero horizontal axis represent the cumulative negative emission amount between 2025 and 2060. (c) indicates the cumulative negative emission amount affected by start-to-retrofit time.

annual negative emission (Figure 5a). For example, BE80_T100 represents 80% of biomass resource potential in the B+T+ scenario, while the maximum co-firing ratio can reach 100%, and the notation of the remaining scenarios can be deduced by analogy. The area between the curve and horizontal axis represents the cumulative negative emission of these 11 scenarios from 2020 to 2060 (Figure 5b). The result shows that if coal power units start and finish retrofit in 2025, the cumulative negative emission up to 2060 could reach as large as 1.49–10.32 Gt, depending on the amount of biomass resource potential and the level of technology breakthrough. Figure 5c shows the relationship between the cumulative negative emission amount and the time to begin retrofit. If we postpone the retrofitting action, the negative emission potential would decrease at a rate of 0.06–0.35 Gt (or 3–4%) per year, implying the urgency to retrofit the current units as soon as possible.

DISCUSSION

To gradually phase down coal in China's coal power generation sector and avoid many side-effects, this study

explored the negative emission potential and the corresponding cost of BCP-CCS. For each of the 4536 coal power units, we have considered their spatial distribution and differentiated characteristics (such as installed capacity, remaining operating life, nearby resources' availability, and transportation distance). Also, the information will help each unit identify the optimal retrofitting plan to achieve the maximum emission reduction. Sensitivity analysis has also been carried out to explore the influence of three key factors (the amount of biomass resource potential, the maximum co-firing ratio of biomass with coal, and the time to start and finish co-firing retrofitting) on the emission reduction or negative emission potential in the coal power generation sector.

Our results show that if we fully exploit China's biomass resources and have a technology breakthrough in the co-firing ratio, China's coal power units would be able to achieve 0.4 Gt of negative emission in 2025 and 10.3 Gt of cumulative negative emission from 2025 to 2060 (equivalent to China's CO₂ emission in 1 year). If we delay the retrofit action, the cumulative negative emission potential in this sector would be reduced at a rate of 0.35 Gt per year. According to the

sensitivity analysis, to achieve negative emission potential, the supply amount of biomass resources should exceed 26.6 EJ (equivalent to 1.65 times the total amount of existing agricultural and forestry residues), and the co-firing ratio should improve from the current 20% to at least 70%. The cost analysis shows that under the current carbon price in China, almost no coal power unit would choose BCP-CCS. Under the European carbon price, the maximum emission reduction amount can reach 1.97 Gt.

In fact, the negative emission potential and the economic cost of BCP-CCS are a complex problem, which involves many topics, such as biomass resource supply, biomass and coal co-firing technology, carbon capture, storage, and even utilization technology, logistics and the transportation process, electricity price, carbon market, and so on. It is difficult to consider all factors in one model and article, which awaits more model coupling and interdisciplinary studies to answer. This article has simplified many factors, which can be improved in the following aspects; First, the cost of R&D technology has great uncertainties, which may influence the cost of CCS technology. Second, CCS technology will cause extra energy consumption, which may increase the CO₂ emission.^{61–63} The state-of-the-art solvent and energy gradient utilization may decrease energy penalty.^{18,64} In addition, the utilization of the captured carbon dioxide for chemicals and materials could decrease the cost of BCP-CCS.⁶⁵ Third, the future population and food consumption will influence the land use spatial pattern,⁶⁶ which may affect the spatial distribution of biomass resources and total resource supply. Fourth, pricing of carbon will significantly change the role of BCP-CCS.⁶⁷ However, China's carbon market is still in the development stage. The setting of carbon prices at present and the dynamic changes in the future may influence the decision of each coal power unit annually and further influence the cumulative emission reduction potential. In the end, China's biomass supply market is not mature yet. The combination of multiple transportation modes and even transnational cooperation may reduce the cost of biomass supply.^{68,69}

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.2c06004>.

Detailed parameters and mathematic formation of LCOE and life cycle CO₂ emission; literature review of biomass resource potential and the comparison with our results; China's coal power units' information and spatial distribution; source and sink matching between coal power units and CCS storage sites; emission reduction results of coal power units; and result data and the codes used for calculation and visualization of emission reduction and economic cost available on the GitHub repository: <https://github.com/Rui-W-A/negative-emission-BCPCCS> (PDF)

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